

Residual Reinforcement Learning for Adaptive Grasp Recovery under Dynamics and Grasp Uncertainty

CS5478 Intelligent Robots: Algorithms and Systems – Li Jiaqi

Motivation and Research Question

Robotic manipulation requires not only establishing a grasp, but also maintaining grasp stability during subsequent motion. A conventional analytical controller can efficiently generate reaching, grasping, and lifting motions, but grasp stability is sensitive to uncertain physical conditions. Variations in object mass and contact friction, together with imperfect initial grasp placement, can cause an initially successful grasp to deteriorate during lifting through slipping, lateral displacement, or rotational instability.

A fixed controller may respond poorly to these changing interaction conditions. Insufficient gripping force may allow a heavy or low-friction object to slip, while continuing an aggressive lifting trajectory may further destabilize the grasp. Conversely, conservative strategies such as continuously applying excessive gripping force or lifting very slowly may improve robustness at the cost of efficiency.

This project investigates whether reinforcement learning can be used specifically as an **online recovery controller**. Normal manipulation is handled by a conventional Cartesian controller. When measurable interaction signals indicate that the grasp has begun to deteriorate, a Soft Actor-Critic (SAC) policy is activated and produces bounded residual corrections to the nominal end-effector motion and gripping force. Once the grasp becomes stable again, control returns to the nominal controller.

Research Question:

How can a recovery-specific SAC policy adapt its corrective actions to different grasp-deterioration states under uncertainty in object mass, friction, and grasp placement?

The objective is not to estimate hidden physical parameters such as mass or friction. Instead, SAC learns how to respond to their observable physical consequences during grasp deterioration. The residual formulation is used only inside the recovery mode: the nominal controller provides the task-level motion prior, while SAC receives bounded authority to modify that motion only when adaptation is required. This mode-switched design concentrates learning on recovery and limits unnecessary learned intervention during stable manipulation.

Task and Uncertainty Model

A 7-DoF Panda robotic arm with a parallel-jaw gripper will be simulated in MuJoCo. The task consists of approaching a cylindrical object, grasping it, and lifting it to a target height.

A **single fixed cylinder geometry** will be used throughout the core experiments. This avoids the substantially harder problem of cross-object generalization and provides a controlled environment for studying grasp recovery.

Three sources of uncertainty will be introduced:

- **Object mass:** cylinder mass m is randomized across episodes.
- **Contact friction:** finger–object friction coefficient μ is randomized across episodes.
- **Grasp placement:** the commanded grasp position is perturbed from the ideal centered grasp by a bounded offset δp_{grasp} .

The grasp location is

$$p_{\text{grasp}} = p_{\text{ideal}} + \delta p_{\text{grasp}},$$

where

$$\delta p_{\text{grasp}} = [\delta x, \delta y, \delta z].$$

Off-center contact can introduce asymmetric loading, lateral motion, and rotational instability. Together with mass and friction variations, this produces several controlled modes of grasp deterioration while keeping object geometry fixed.

The values of m and μ are not included in the policy observation. SAC must adapt through observable robot–object interaction rather than direct knowledge of the randomized dynamics parameters.

Nominal Manipulation and Recovery Architecture

A conventional controller performs the ordinary manipulation task. A finite-state machine executes

APPROACH → DESCEND → CLOSE → LIFT.

A Jacobian-based Cartesian controller tracks the desired end-effector trajectory and performs nominal grasping and lifting.

During lifting, a deterioration measure D_t is computed using ground-truth interaction state available from MuJoCo. When deterioration exceeds a threshold,

$$D_t > D_{\text{enter}},$$

the system transitions from nominal lifting to

RECOVERY

.

During recovery, SAC generates residual corrections to the nominal controller.

When grasp stability has been restored,

$$D_t < D_{\text{exit}},$$

for a predefined number of consecutive control steps, the system returns to nominal lifting.

Two thresholds are used with

$$D_{\text{exit}} < D_{\text{enter}}$$

to provide hysteresis and prevent rapid switching between nominal and recovery control.

The resulting control architecture is

$$\text{Nominal Manipulation} \xrightarrow{\text{deterioration detected}} \text{SAC Recovery} \xrightarrow{\text{grasp stabilized}} \text{Nominal Manipulation}.$$

A complete object drop terminates the episode. Regrasp planning after complete failure is outside the project scope.

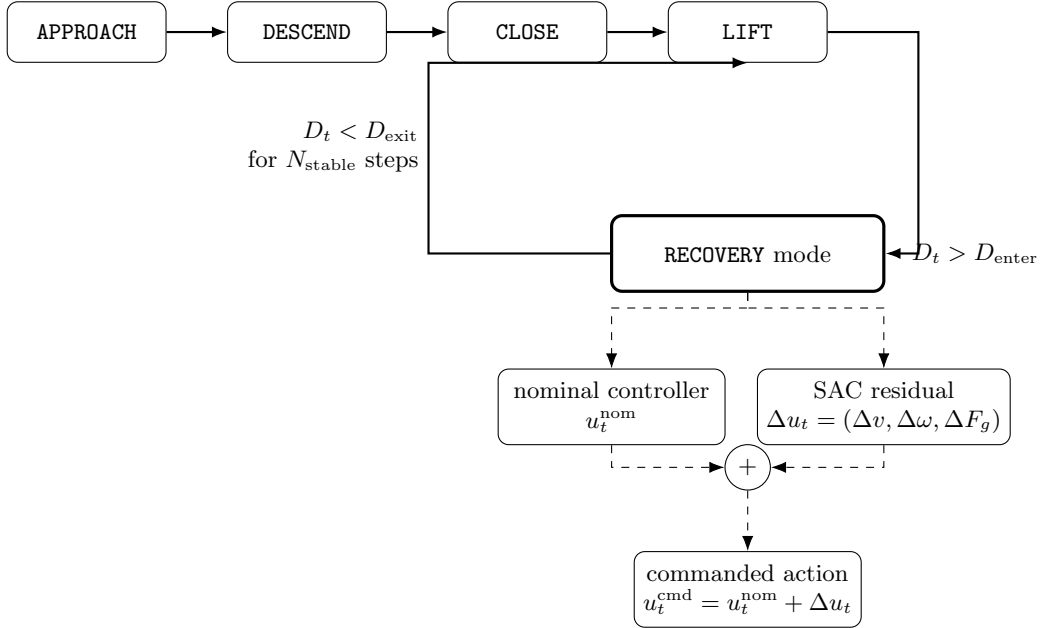


Figure 1: Mode-switched recovery architecture. The nominal finite-state machine executes approach, descend, close, and lift. When the deterioration measure exceeds the entry threshold ($D_t > D_{\text{enter}}$), control switches to **RECOVERY** mode. There, a Soft Actor-Critic policy receives the ground-truth interaction state and outputs bounded residuals $(\Delta v, \Delta \omega, \Delta F_g)$, which are added to the nominal command u_t^{nom} . When $D_t < D_{\text{exit}}$ for N_{stable} consecutive control steps, control returns to nominal lifting. Solid arrows denote FSM transitions; dashed arrows denote signal flow.

Deterioration Detection

Grasp deterioration will be detected using ground-truth state information available directly from MuJoCo. The project focuses on recovery control rather than perception or slip-detection uncertainty, so deterioration detection does not need to be learned.

Let

$$p_{\text{rel},t} = p_{\text{object},t} - p_{\text{gripper},t},$$

and

$$v_{\text{rel},t} = v_{\text{object},t} - v_{\text{gripper},t}.$$

A reference relative pose is recorded after the grasp has been established. A normalized deterioration measure will then be constructed from deviations from this stable reference:

$$D_t = w_p \frac{\|\Delta p_{\text{rel},t}\|}{p_{\text{scale}}} + w_v \frac{\|v_{\text{rel},t}\|}{v_{\text{scale}}} + w_\theta \frac{|\theta_{\text{rel},t}|}{\theta_{\text{scale}}} + w_\omega \frac{\|\omega_{\text{rel},t}\|}{\omega_{\text{scale}}} + w_c C_t.$$

Here, $\Delta p_{\text{rel},t}$ is the displacement from the established grasp configuration and $\theta_{\text{rel},t}$ represents relative orientation deviation. For the parallel-jaw gripper, the contact term is defined as

$$C_t = \left| \frac{f_{L,t} - f_{R,t}}{f_{L,t} + f_{R,t} + \epsilon} \right| + 1[\text{either fingertip loses contact}],$$

where $f_{L,t}$ and $f_{R,t}$ are the normal contact-force magnitudes at the left and right fingertips and $\epsilon > 0$ prevents division by zero. Thus, C_t increases under asymmetric contact loading and receives an additional penalty when either fingertip loses contact.

The scale factors make the components of D_t dimensionless so that quantities with different physical units do not dominate purely because of numerical scale. Their numerical values, together with $w_p, w_v, w_\theta, w_\omega, w_c$, will be calibrated using nominal and deteriorating trajectories from a small validation set rather than fixed before observing the simulated interaction data.

Larger D_t represents more severe grasp deterioration. Its finite-difference trend

$$\dot{D}_t \approx \frac{D_t - D_{t-1}}{\Delta t}$$

where Δt is the interval between consecutive policy/control decisions. Thus, $\dot{D}_t$ is computed at the same frequency at which the recovery policy observes the state and produces a residual action, rather than at the potentially higher MuJoCo simulation frequency. This trend will also be available to the recovery controller, providing an explicit indication of whether the grasp is improving or worsening without requiring recurrent state estimation or observation-history modeling.

When

$$D_t > D_{\text{enter}},$$

the controller enters the **RECOVERY** state. Recovery is considered successful when

$$D_t < D_{\text{exit}}$$

for N_{stable} consecutive control steps. The thresholds satisfy

$$D_{\text{exit}} < D_{\text{enter}}$$

to provide hysteresis. $D_{\text{enter}}, D_{\text{exit}}$, and N_{stable} will be calibrated on a small validation set and then held fixed across all controllers and evaluation conditions.

The same ground-truth deterioration detector will be used for all recovery methods. This removes perception and detection accuracy as experimental variables and isolates the central question of **how to recover after grasp deterioration occurs**.

SAC Recovery Controller

Soft Actor-Critic is activated only in the **RECOVERY** state.

The policy produces bounded Cartesian velocity and gripping-force residuals:

$$a_t = [\Delta v_x, \Delta v_y, \Delta v_z, \Delta \omega_x, \Delta \omega_y, \Delta \omega_z, \Delta F_g].$$

During recovery,

$$\begin{bmatrix} v_t^{\text{cmd}} \\ \omega_t^{\text{cmd}} \end{bmatrix} = \begin{bmatrix} v_t^{\text{nominal}} \\ \omega_t^{\text{nominal}} \end{bmatrix} + \begin{bmatrix} \Delta v_t \\ \Delta \omega_t \end{bmatrix},$$

and

$$F_{g,t}^{\text{cmd}} = F_{g,t}^{\text{nominal}} + \Delta F_{g,t}.$$

Velocity residuals allow SAC to make interventions that cannot be represented by a small fixed pose offset. For example, if

$$\Delta v_z < -v_z^{\text{nominal}},$$

then

$$v_z^{\text{cmd}} < 0,$$

allowing the policy to temporarily reverse the lifting motion if such a strong counter-action improves recovery.

Possible learned responses include:

- increasing gripping force,
- reducing lifting velocity,
- temporarily pausing or reversing lifting motion,
- following lateral object displacement,
- applying orientation corrections to counter rotational instability.

The project does not assume that these behaviors are optimal. SAC is given sufficient residual authority to explore them, and the learned behavior will be analyzed experimentally. Residual action bounds will be selected from the validated nominal controller so that corrective motion remains physically reasonable while still permitting lift reversal. Action-rate limits will additionally prevent unrealistically abrupt changes. The exact bounds will be fixed before comparative evaluation and shared across all learned-policy experiments.

Recovery Observations

During the **RECOVERY** state, SAC receives directly observable simulator state and interaction information relevant to recovery, including:

- joint positions q and velocities $\dot{q}$,
- end-effector pose and Cartesian velocity,
- object pose and velocity,
- object–gripper relative position and velocity,
- object orientation and angular velocity,
- fingertip contact and force information,
- current nominal Cartesian command,
- deterioration measure D_t and deterioration trend $\dot{D}_t$.

These quantities are obtained directly from MuJoCo ground-truth simulation state. The project therefore does not address perception uncertainty or learned state estimation.

The randomized physical parameters

$$m, \quad \mu$$

are deliberately excluded from the SAC observation. The policy must learn recovery actions from the resulting physical interaction rather than being explicitly told the object’s mass or friction coefficient. These parameters are not completely unobservable: their effects are reflected indirectly in motion and contact forces, so SAC may implicitly adapt to them through interaction feedback without an explicit system-identification module.

Thus, the control problem is

$$\boxed{\text{ground-truth interaction state} \rightarrow \text{SAC recovery action}}$$

rather than a combined perception, parameter-estimation, and control problem.

Explicit latent-state learning, observation-history modeling, system identification, and estimation of mass or friction are outside the project scope.

Recovery-Focused Training

SAC will be trained primarily on **deteriorating grasp states**, rather than throughout the complete manipulation trajectory.

First, the nominal controller will be executed under randomized combinations of

$$m, \quad \mu, \quad \delta p_{\text{grasp}}.$$

When

$$D_t > D_{\text{enter}},$$

the corresponding physically valid MuJoCo state can be used as the beginning of a recovery episode.

The resulting training distribution will contain a range of deterioration conditions, including mild, moderate, and severe cases.

Conceptually,

$$\mathcal{B}_{\text{recovery}} = \{s_{\text{mild}}, s_{\text{moderate}}, s_{\text{severe}}\}.$$

Recovery training then follows

$$s_{\text{deteriorating}} \xrightarrow{\text{SAC}} \begin{cases} \text{grasp stabilized,} \\ \text{continued deterioration,} \\ \text{object dropped.} \end{cases}$$

This concentrates reinforcement-learning samples on the part of the task where learned adaptation is required. SAC therefore receives many opportunities to explore different corrective actions, including strong counter-residual motions, rather than spending most training steps performing ordinary stable manipulation.

For the core experiment, recovery initial states will be stratified by the initial normalized deterioration value D_t so that mild, moderate, and severe but recoverable states are represented in the training distribution.

A curriculum

$$\text{mild} \rightarrow \text{moderate} \rightarrow \text{severe}$$

is an **optional training extension**, used only if direct training on the stratified distribution fails to learn reliably from severe recovery states. If used, it begins with lower- D_t recoverable states and introduces more severe states after recovery performance becomes stable. The core experimental claim therefore does not depend on a particular curriculum schedule.

A recovery-training episode terminates when one of the following occurs:

- **successful stabilization:** $D_t < D_{\text{exit}}$ for N_{stable} consecutive control steps;
- **object drop:** the grasp is completely lost;
- **unrecoverable state:** the object leaves the valid manipulation workspace or recoverable contact configuration;
- **timeout:** a fixed maximum recovery duration is reached.

Reward Design

Because SAC operates only during recovery, the reward can focus directly on restoring grasp stability. To provide dense recovery feedback without directly rewarding arbitrary cycles in D_t , deterioration progress will be incorporated using potential-based shaping.

Define the potential

$$\Phi(s_t) = -\lambda_D D_t.$$

The shaping reward is

$$r_t^{\text{shape}} = \gamma \Phi(s_{t+1}) - \Phi(s_t).$$

The complete conceptual reward is then

$$r_t = r_t^{\text{shape}} - \lambda_f F_{g,t}^2 - \lambda_c (\|\Delta v_t\|^2 + \|\Delta \omega_t\|^2) - \lambda_t + 1[\text{recovery succeeds at } t] r_{\text{recovered}} - 1[\text{object drops at } t] r_{\text{drop}}.$$

The potential term provides dense feedback when the policy moves toward lower deterioration while retaining a principled shaping structure. The gripping-force penalty discourages continuous maximum squeezing, the residual penalty discourages unnecessarily aggressive motion, and the time penalty encourages rapid recovery.

The recovery and drop terms are zero on ordinary intermediate steps. Successful stabilization receives the terminal bonus $r_{\text{recovered}}$, while an object drop receives the terminal penalty r_{drop} . Reward weights will be tuned on a small validation set and then fixed for the main experiments. In particular, λ_c will be chosen so that a successful recovery requiring a strong corrective action remains preferable to failure with negligible residual action.

The desired behavior is therefore not necessarily small residual control, but rather

the minimum intervention sufficient for successful recovery.

Training Stages

Development will proceed incrementally.

Stage 1 – Nominal Manipulation Controller

Implement and validate stable approach, grasp, and lift behavior under nominal conditions.

Stage 2 – Deterioration Generation and Detection

Introduce controlled mass, friction, and grasp-offset variations. Define and validate D_t and the transition into RECOVERY.

Stage 3 – Low-Dimensional Recovery

Initially train SAC using

$$a_t = [\Delta v_z, \Delta F_g].$$

This experiment tests whether SAC can learn the fundamental trade-off between modifying lifting motion and increasing gripping force, including whether severe deterioration produces strong counter-motion.

Stage 4 – Cartesian Recovery

If grasp-offset experiments generate meaningful lateral or rotational failure modes, expand the action space to

$$a_t = [\Delta v_x, \Delta v_y, \Delta v_z, \Delta \omega_x, \Delta \omega_y, \Delta \omega_z, \Delta F_g].$$

This staged design prevents unnecessary action-space complexity while allowing richer recovery behavior when justified by the observed physics.

Stage 5 – Randomized Recovery Training

Train across randomized m , μ , and δp_{grasp} and evaluate on held-out combinations.

Baselines

Three controllers will be evaluated.

1. Nominal Controller

The nominal controller continues lifting without a dedicated recovery mechanism.

2. Hand-Designed Reactive Recovery

The same deterioration detector activates a stronger fixed feedback controller using both deterioration severity and its trend. A representative form is

$$\Delta F_g = k_{p,F} D_t + k_{d,F} \dot{D}_t,$$

and

$$\Delta v_z = -\text{clip}\left(k_{p,v} D_t + k_{d,v} \dot{D}_t, 0, \Delta v_{z,\text{max}}\right).$$

The gains will be tuned on the same validation conditions used for detector calibration and then held fixed during evaluation. For a fair comparison, the heuristic controller and SAC will use the **same residual action bounds and action-rate limits**. Thus, neither method receives greater corrective authority than the other. This baseline can react continuously to worsening deterioration rather than applying only a binary recovery rule.

3. SAC Recovery

The same deterioration detector activates the learned residual recovery policy.

Thus, the heuristic and SAC methods receive the same deterioration trigger and ground-truth interaction information. Their primary difference is the recovery action-selection strategy.

This provides a stronger comparison than evaluating SAC only against a controller with no recovery behavior.

Experimental Evaluation

The controllers will be evaluated on identical deterioration conditions generated from combinations of

$$(m, \mu, \delta p_{\text{grasp}}).$$

Experiment	Comparison	Metric	Purpose
Recovery	Nominal, heuristic, SAC	Recovery rate	Main comparison
Severity	Mild, moderate, severe	Recovery vs. D_t	Test intervention strength
Mass/Friction	Held-out m, μ	Recovery degradation	Dynamics robustness
Grasp Offset	Increasing δp	Recovery vs. offset	Placement robustness
Action Space	2D vs. Cartesian residual	Recovery rate, time, residual effort	Value of richer corrections
Contact Feedback	With/without contact signals	Recovery rate	Feedback ablation

The primary metric is

$$\text{Recovery Rate} = \frac{N_{\text{deteriorated grasps successfully stabilized}}}{N_{\text{recovery attempts}}}.$$

Additional metrics include:

- object-drop rate,
- maximum relative slip,
- time-to-recovery,
- gripping effort,
- residual control magnitude,
- subsequent task-completion rate.

The learned SAC actions will also be analyzed as a function of initial deterioration severity. Recovery rate will be reported across bins of initial D_t , allowing direct comparison of mild, moderate, and severe recovery conditions. In particular, the experiments will examine whether stronger deterioration produces stronger corrective actions and whether SAC learns temporary counter-motion rather than relying only on increased gripping force.

Recovery-rate estimates will be reported with confidence intervals over repeated evaluation episodes. Representative trajectories of D_t , F_g , v_z , and Δv_z will also be plotted for the heuristic and SAC controllers to show how their recovery behavior differs over time. Failed recovery episodes will be categorized qualitatively, for example as insufficient intervention, destabilizing intervention, or states that are already physically unrecoverable.

For the contact-feedback ablation, contact-derived features will be removed from the SAC observation while the ground-truth deterioration detector itself remains unchanged. This preserves the same recovery trigger and success criterion across the ablation.

Generalization Scope

The project does not aim to learn a universal grasp-recovery policy.

Generalization is restricted to a fixed cylindrical object geometry with varying

$$m, \quad \mu, \quad \delta p_{\text{grasp}}.$$

Training will use sampled combinations from predefined ranges. Evaluation will reserve **held-out regions of the joint uncertainty space** within the overall experimental ranges. Selected intervals or combinations of m , μ , and δp_{grasp} will therefore be excluded from training and used only for evaluation. This is stronger than drawing a different random sample from the same training distribution while avoiding a broad out-of-range extrapolation claim.

The intended question is whether the learned recovery policy adapts to **unseen regions formed by familiar uncertainty factors**, rather than whether it transfers to arbitrary object geometries.

Cross-object geometry generalization is outside the core scope.

Expected Contribution

Online slip detection, reactive grasp stabilization, and residual reinforcement learning have been studied previously. This project therefore does not claim the general concepts of grasp recovery or residual control as novel.

The project instead contributes a focused empirical study of **mode-switched residual SAC for grasp recovery**. Specifically, it investigates:

- a recovery-focused training methodology that concentrates reinforcement-learning samples on rare but important deteriorating grasp states,
- the conditions under which learned residual recovery improves upon a tuned fixed reactive recovery controller under controlled mass, friction, and grasp-placement uncertainty,
- the corrective behaviors SAC discovers as deterioration severity increases, including whether it learns stronger grip adjustment, slowing, pausing, or temporary counter-motion.

The contribution is therefore not dependent on SAC universally outperforming the heuristic. Characterizing where learned recovery helps, where a fixed controller is sufficient, and how the learned policy responds to different deterioration regimes is itself a meaningful experimental result.

Scope and Risk Management

The minimum complete project consists of:

- one cylindrical object,
- a stable nominal Cartesian grasp-and-lift controller,
- randomized mass and friction,
- controlled grasp-placement offsets,
- a ground-truth deterioration detector,
- recovery-focused SAC training,
- a low-dimensional recovery action space,
- nominal and heuristic recovery baselines,
- evaluation on held-out uncertainty combinations.

The full Cartesian residual action space is an extension built on top of the low-dimensional recovery experiment. If lateral and rotational corrections do not provide meaningful benefits for the simulated failure modes, the project remains complete with the simpler recovery action space.

Explicit latent-state learning, system identification, estimation of mass or friction, perception uncertainty, cross-object generalization, RGB-D perception, autonomous grasp planning, and regrasping after complete object drop are outside scope.

This design keeps the learning problem focused: the conventional controller handles ordinary manipulation, the ground-truth detector determines when recovery is required, and reinforcement learning is responsible specifically for learning **how to recover**.

References

- [1] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor,” *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.
- [2] A. Y. Ng, D. Harada, and S. Russell, “Policy Invariance Under Reward Transformations: Theory and Application to Reward Shaping,” *Proceedings of the 16th International Conference on Machine Learning (ICML)*, 1999.
- [3] T. Johannink et al., “Residual Reinforcement Learning for Robot Control,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2019.